Debjyoti Chowbey

Debjyoti Chowbey | +91 75869 23485 | deb7741 | Debjyoti Chowbey

Education

Tezpur University Graduated 2025
B.Tech in Electronics and Communication Engineering *GRADE: 8.06*

Publications

D. Chowbey, R.K. Baruah
"Low-Power CMOS Operational Amplifier Design in 45nm Technology," In *Proc. IEEE Devices for Integrated Circuit (DevIC)*, Kalyani, India, Apr. 2025. DOI: 10.1109/DevIC63749.2025.11012572

Experience

Research Assistant (Digital Design) Present
Raman Research Institute — SARAS Project *Supervisor: Dr. Saurabh Singh*

- Working on **SARAS 3**, a precision radio telescope to detect the ultra-faint **21-cm signal from Cosmic Dawn**, by designing and optimizing the **digital spectrometer backend**.

Undergraduate Research Assistant Feb 2024 – May 2025
VLSI Lab, Tezpur University *Cadence Virtuoso, CMOS Design, Mixed-Signal Processing*

- Designed and simulated **Op-Amps, LNAs, and bandgap voltage references** using **Cadence Virtuoso**, optimized for **low power and precision**.
- Developed **high-gain amplifier circuits** for **medical instrumentation**, improving power efficiency by **30%**.
- Served as **TA for CMOS Analog Design Lab**, mentoring **10+ MTech students** in the layout, simulation and verification of the core analog building blocks.

Summer Research Intern June 2024 – July 2024
IEEE AP MTTS SBC – IIT Kharagpur *DSP, MATLAB, Machine Learning, Python*

- Researched **under-sampled signal reconstruction** techniques for **radar and communication systems**, enhancing detection under noise.
- Developed **adaptive filtering algorithms** in **MATLAB** using **MVDR beamforming** to improve **SNR by 20%**.
- Implemented **Fourier analysis, sparse recovery, and ML-based signal enhancement** for real-time DSP.

Projects

Low-Power Operational Amplifier Design in 45 nm Technology Sep. 2024 – May 2025
Cadence Virtuoso, ADE L/XL, Layout XL *Analog IC Design, Post-Layout Simulation, GDSII*

- **Awarded "Best Project"** for Final Year Thesis among 60 students in the Class of 2025 for innovation and performance in analog IC design.
- Designed and optimized a **low-power two-stage Op-Amp** in **45 nm CMOS**, achieving **67 dB gain**, **3.7 MHz GBW**, and **2 V/μs slew rate** under **10 µW**.
- Completed **layout**, performed **DRC**, **LVS**, and used **Quantus** for parasitic extraction to enable **post-layout simulations**.
- Validated robustness with **90 dB CMRR** and **140 dB PSRR**.

Design of an 8-bit ALU using Verilog Apr. 2024 – Aug. 2024
Xilinx ISE, ModelSim, Verilog, FPGA *RTL Design, Simulation, Optimization*

- Developed an **8-bit ALU** supporting **16 operations** (arithmetic, logic, shift, compare) with a **4-bit opcode decoder**.
- Simulated using **Xilinx ISE** and **ModelSim**, verified via waveform analysis.
- Resolved logic synthesis issues and optimized **timing paths** for minimal violations.

CCD Sensor Characterization and Photometric Data Optimization May 2024 – June 2024
MaximDL, Apogee Alta F6 CCD, Python *Sensor Analysis, Image Processing, Automation*

- Characterized **dark current response** of **Apogee Alta F6 CCD**, reducing **thermal noise by 15%** via temperature control.
- Built a **Python-based automation pipeline** to streamline **CCD data calibration and processing**.
- Implemented **hot pixel correction**, generating **dark-frame maps** to improve **photometric accuracy**.

Relevant Coursework

- **Analog Circuits**
- **Digital System Design**
- **VHDL and FPGA Design**
- **Digital Signal Processing**
- **Signals and Systems**
- **Network Theory**
- **Electronic Devices**
- Control Systems
- **Mixed Signal Design**
- Computer Architecture
- MEMS and Microfabrication
- Python for Scientific Computing

Skills

Circuit Design: Op-Amps, ADC/DAC fundamentals, Bandgap References, Data Converters, Biasing, Filters

Device Physics: MOSFET/BJT Operation, Device Modeling

Analog Layout: Matching, Common-Centroid, DRC/LVS, Shielding, Parasitic Handling

HDL & Mixed Signal: Verilog, SystemVerilog, VHDL

Digital Design: FSMs, RTL Design, Timing Closure, Clock Domain Crossing (CDC)

Simulation & Verification: DC Sweep, AC/Noise, Transient, Monte Carlo, Corner Analysis

EDA Tools: Cadence Virtuoso, Spectre, Quantus, Xcelium, Synopsys DC, ModelSim

Scripting: Python, MATLAB, Shell Scripting, SKILL (Cadence)

OS and Tools: Linux (Scripting, Toolchains), Git, Jenkins, Makefiles

Soft Skills: Problem-Solving, Precision, Collaboration, Technical Communication

Leadership, Achievements & Interests

★ Leadership Roles

- Coordinator, Astronomy Club (2022–2024)
  - → Led 200+ participants in night-sky observation and astrophotography sessions.
  - → Initiated a mentoring program for 50+ student members.
  - → Conducted school outreach sessions to promote astronomy interest.
- Member, National Service Scheme (2022–2023)
  - → Helped collect 100+ blood units during donation drives.
  - → Mobilized 150+ volunteers for community clean-up campaigns.
  - → Led STEM outreach in schools, inspiring civic and scientific curiosity.

🏆 Honors & Certifications

- **GATE EC – 2025 Qualified**
- Top **5%** in ECE Department, Class of 2025
- **2nd Place** – IEEE Texplorer Research Fair
- Pre-Finalist – eYantra Robotics, IIT Bombay
- Certified in Data Analysis using Python – IUCAA
- Presented at **IEEE DevIC 2025**, Kalyani

♥ Interests & Hobbies

- Astrophotography: Deep-sky imaging, telescope handling
- Adventure: Cycling, hiking, night walks
- Sports: Cricket, badminton